ETH zürich

D INFK

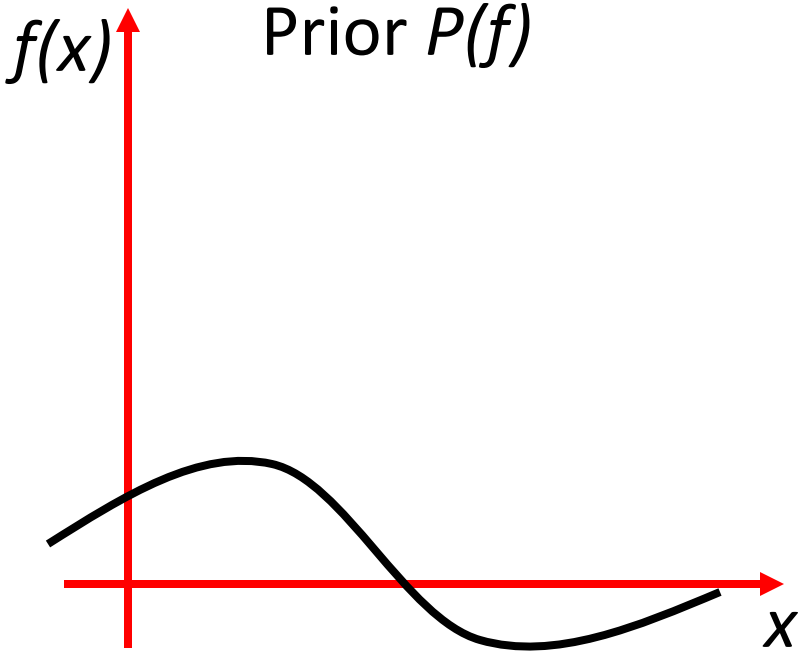
Probabilistic Artificial Intelligence

Gaussian Processes (continued)

Prof. Andreas Krause
Learning and Adaptive Systems (las.ethz.ch)
Institute for Machine Learning

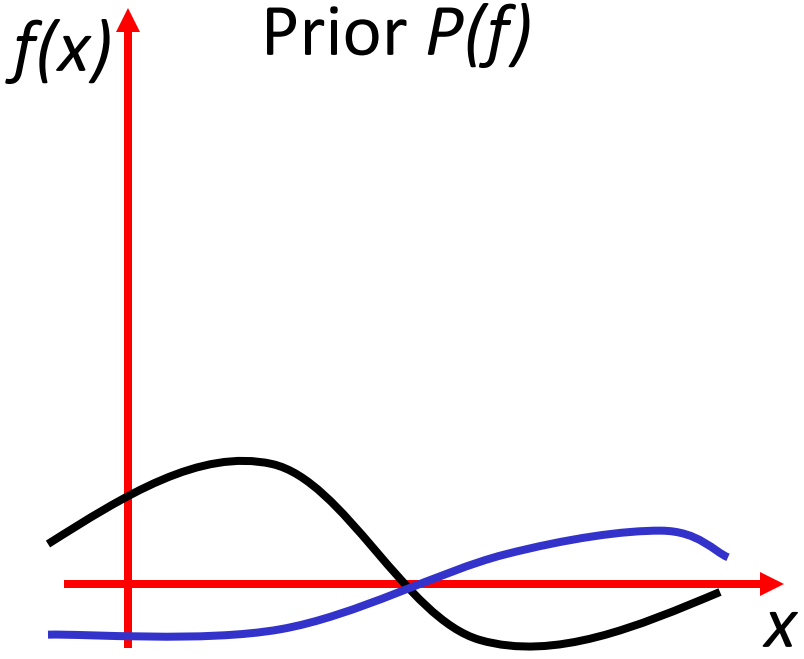
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]



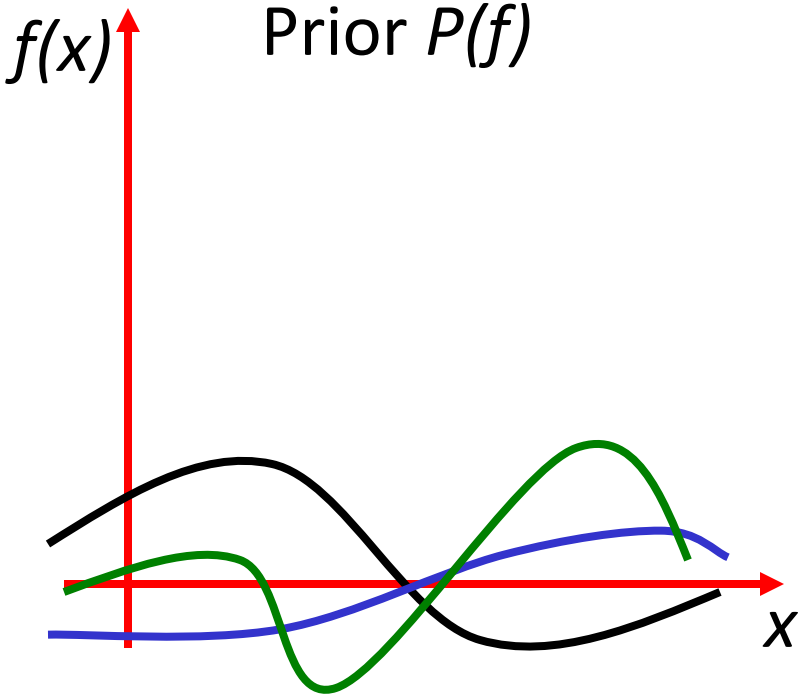
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]



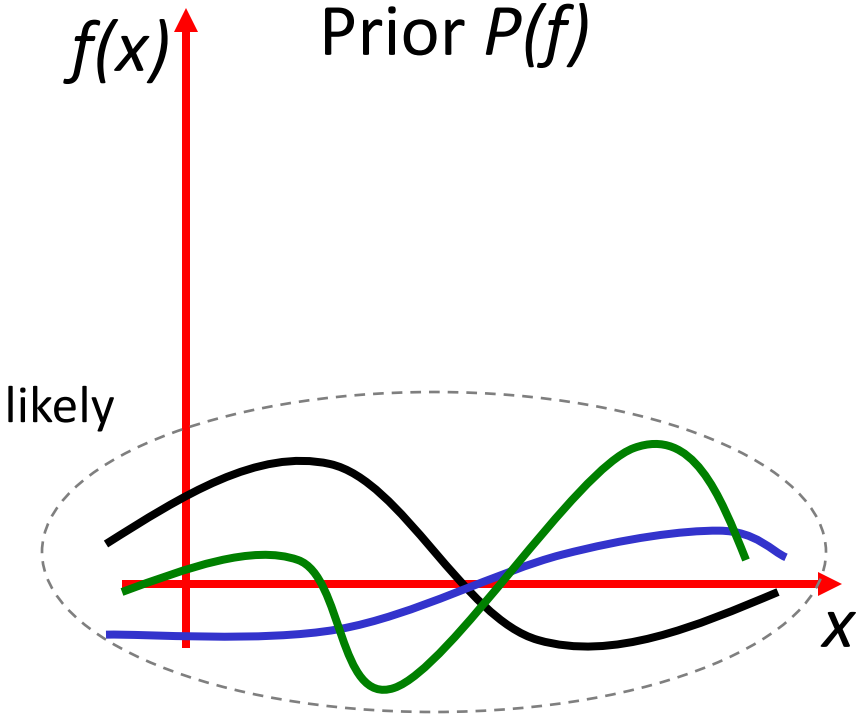
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]



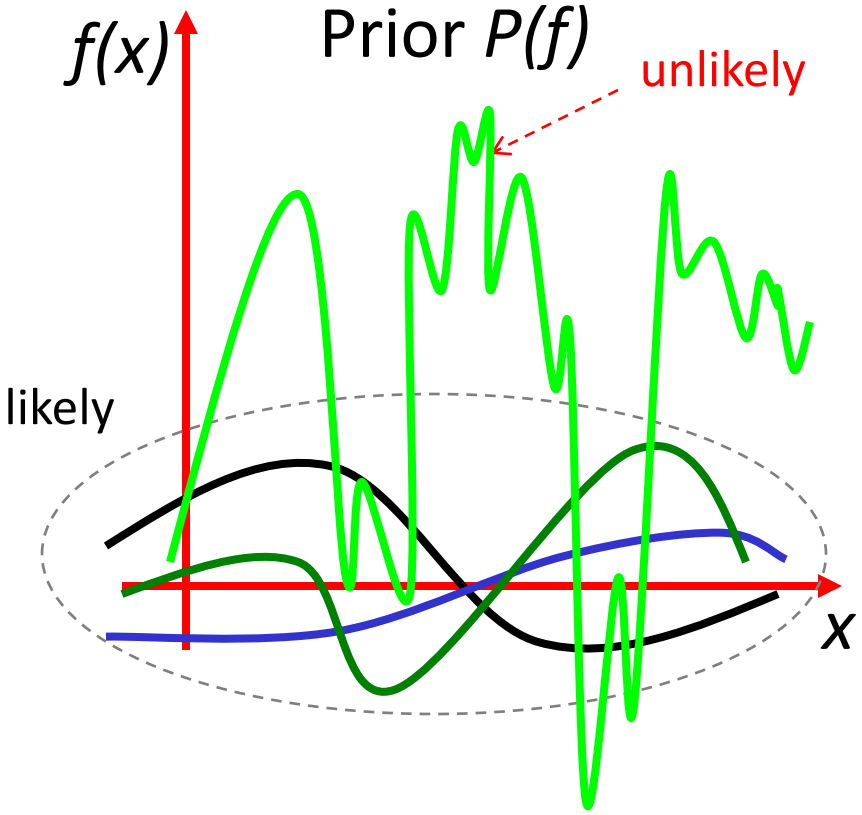
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]



Bayesian learning with Gaussian processes

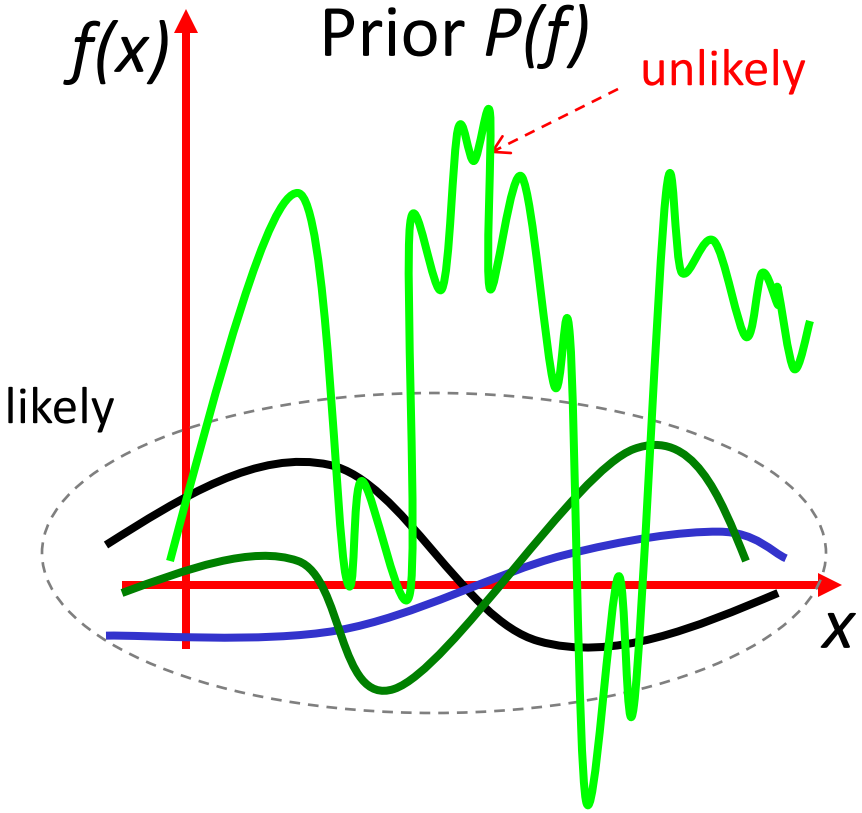
[c.f. Rasmussen & Williams 2006]

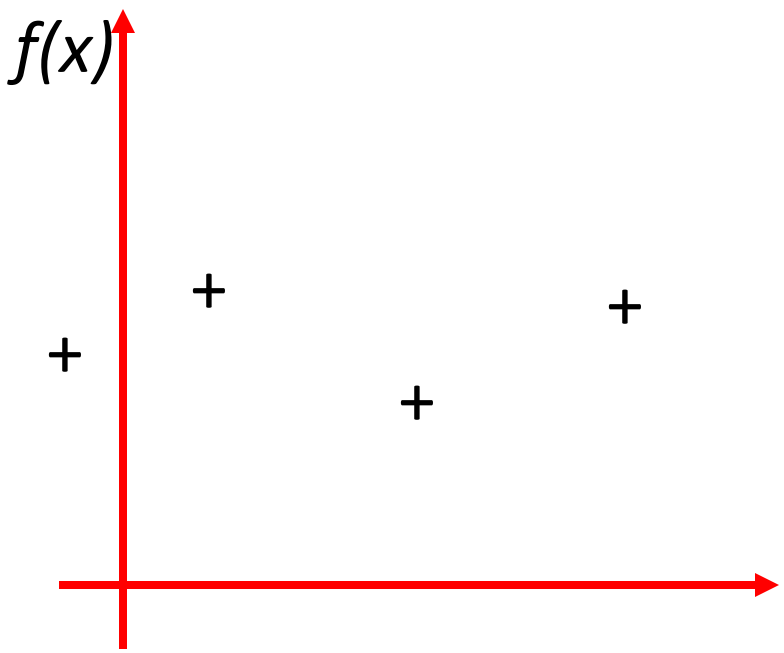


Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]

Likelihood: *P(data | f)*



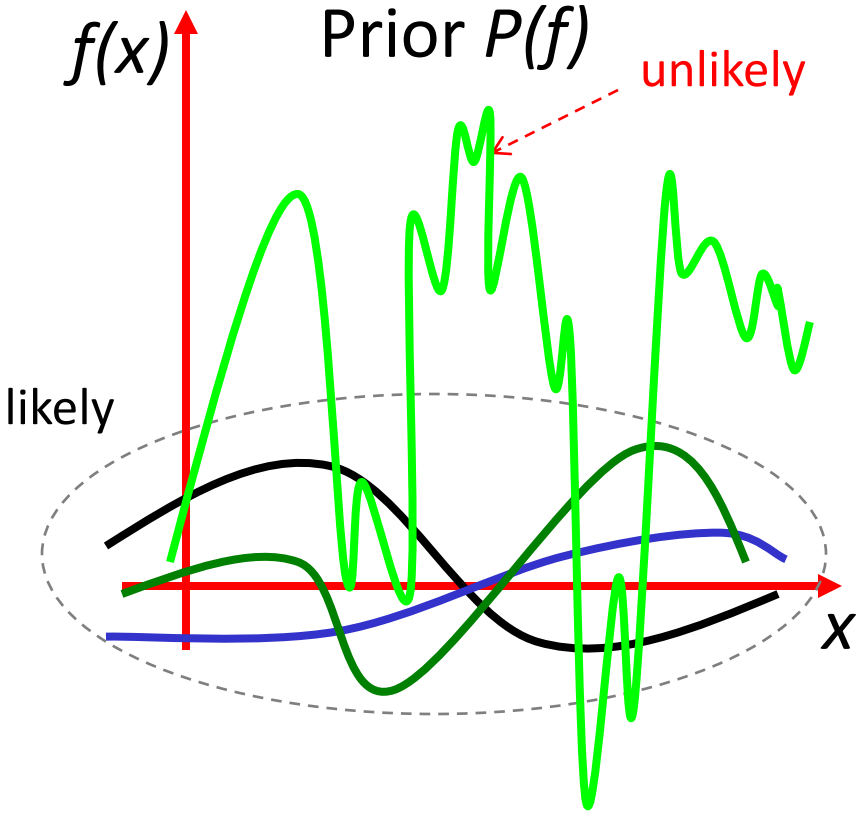


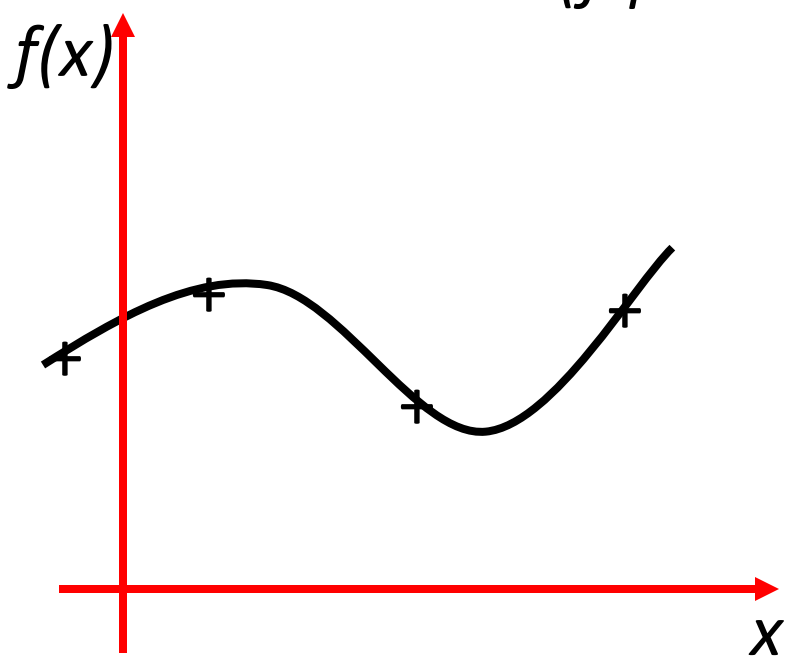
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]

Likelihood: *P(data | f)*

➔ Posterior: *P(f | data)*



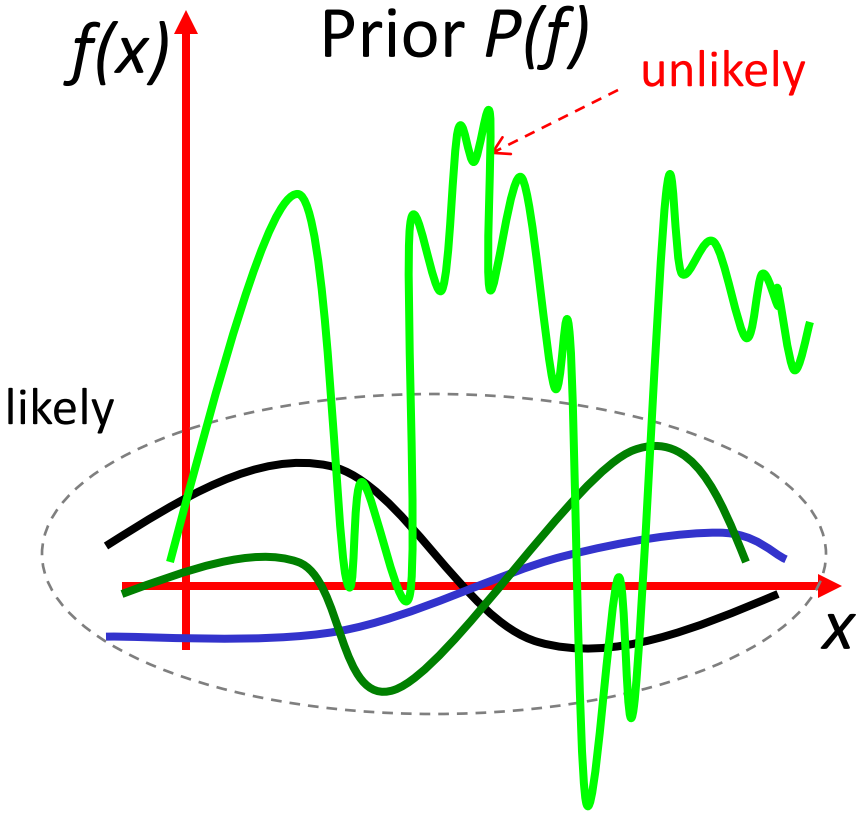


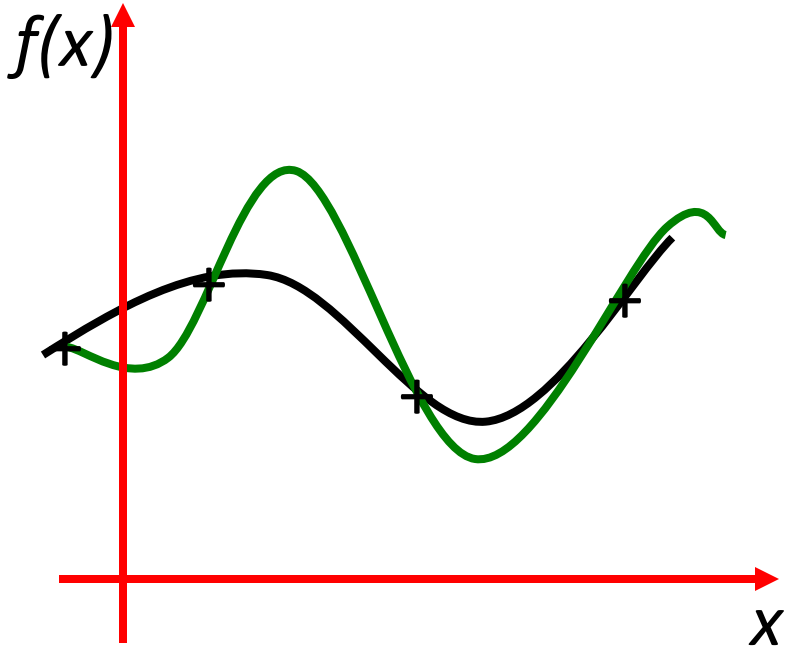
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]

Likelihood: *P(data | f)*

➔ Posterior: *P(f | data)*



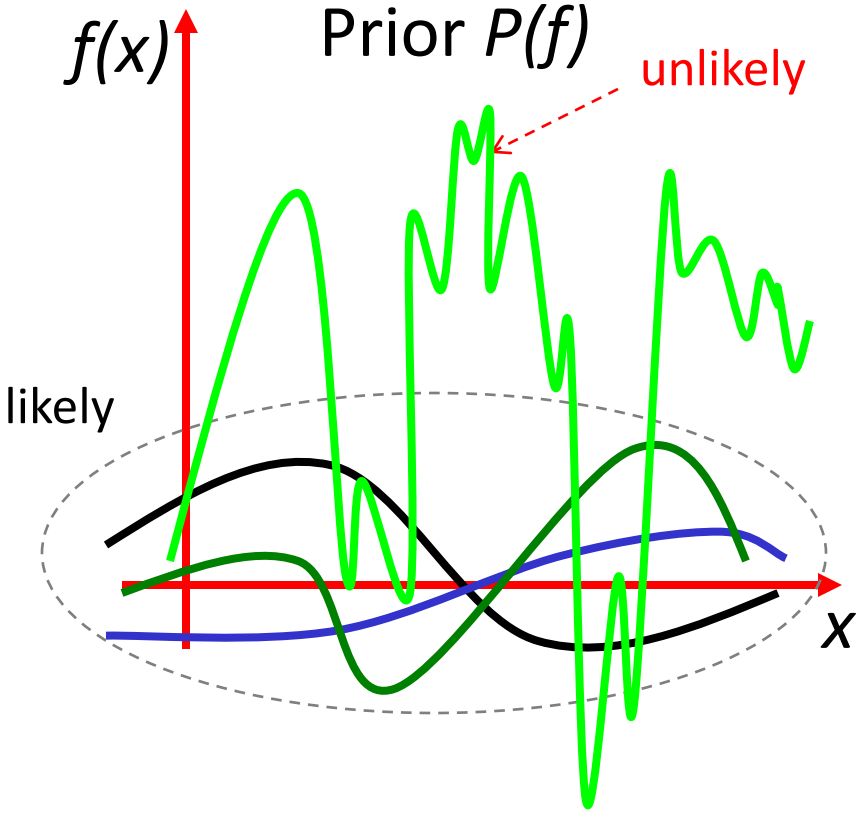


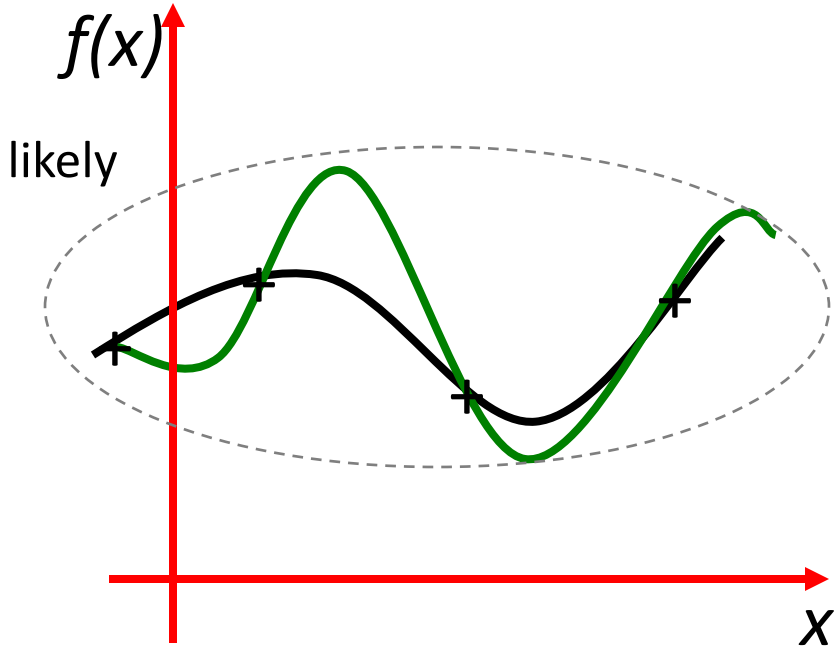
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]

Likelihood: *P(data | f)*

➔ Posterior: *P(f | data)*



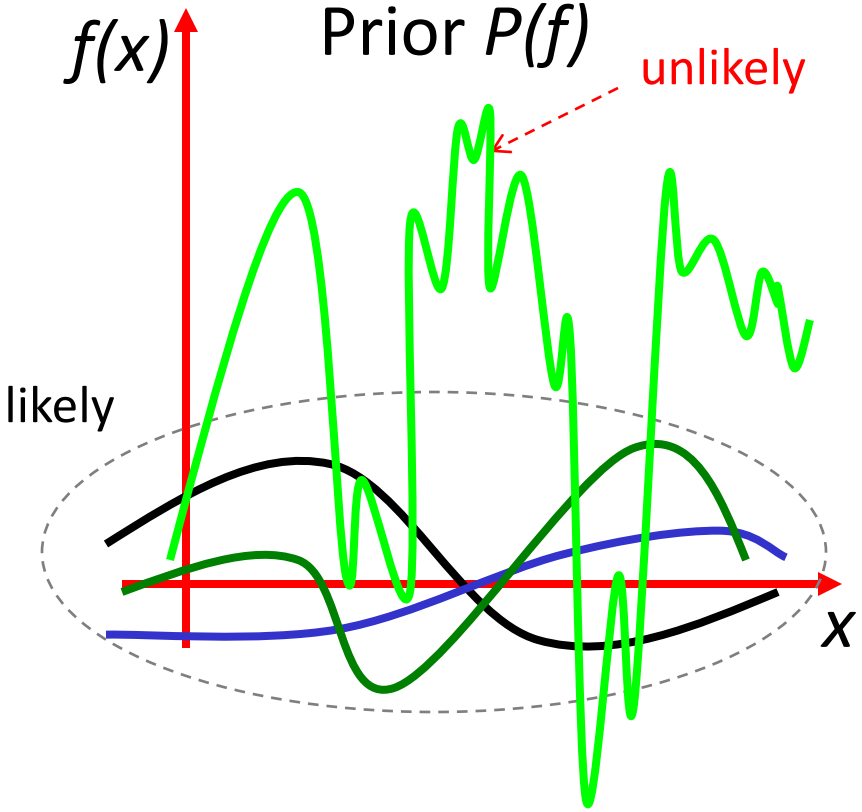


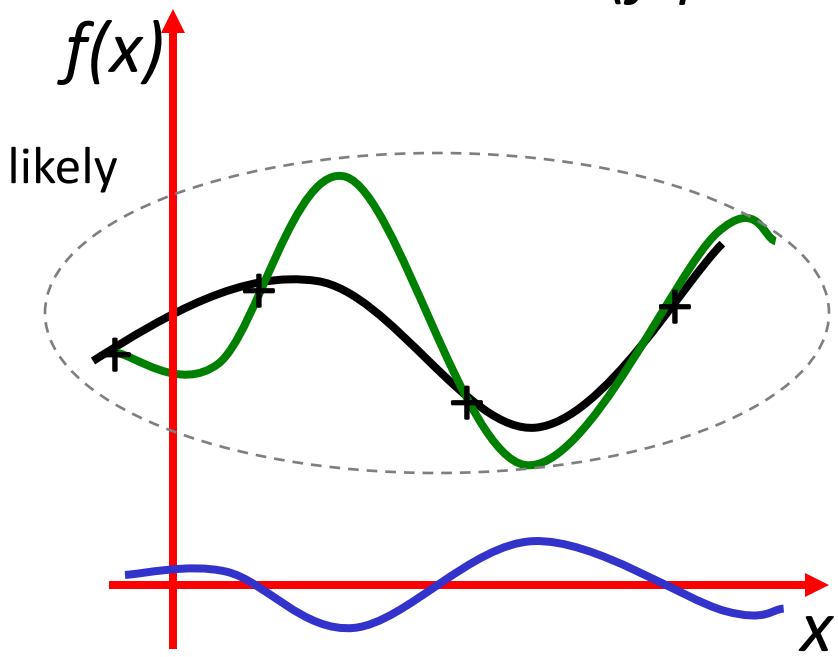
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]

Likelihood: *P(data | f)*

➔ Posterior: *P(f | data)*



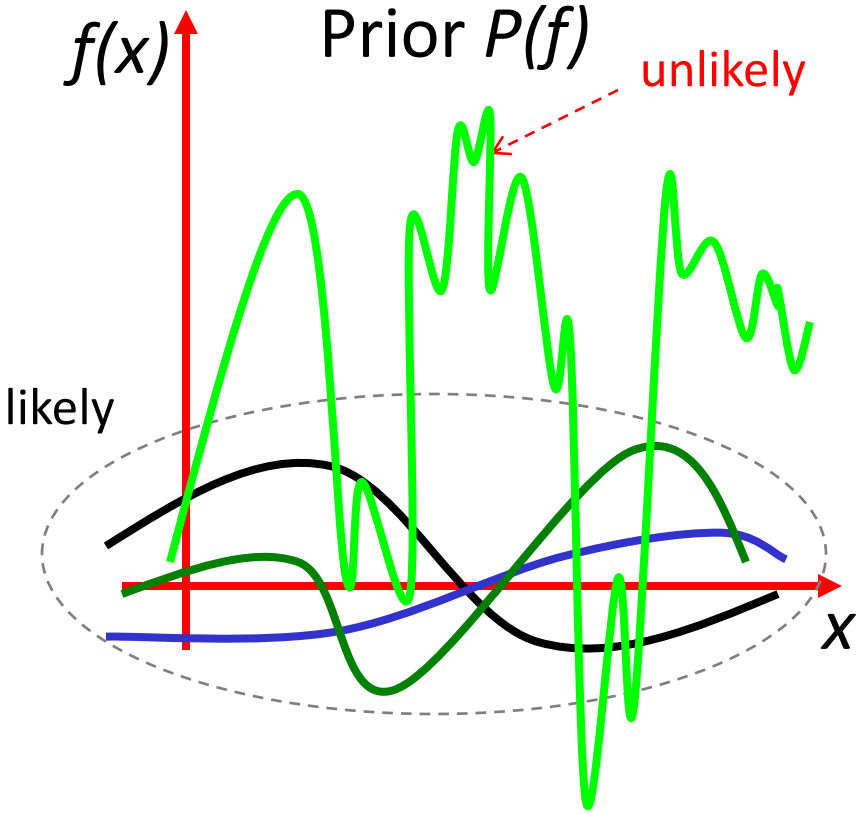


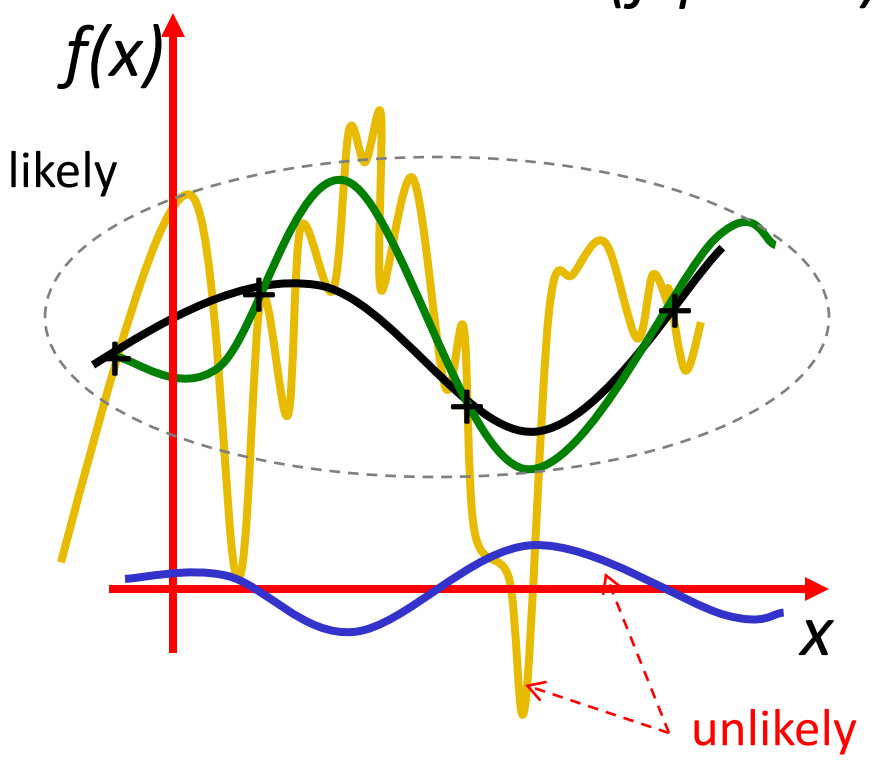
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]

Likelihood: *P(data | f)*

➔ Posterior: *P(f | data)*



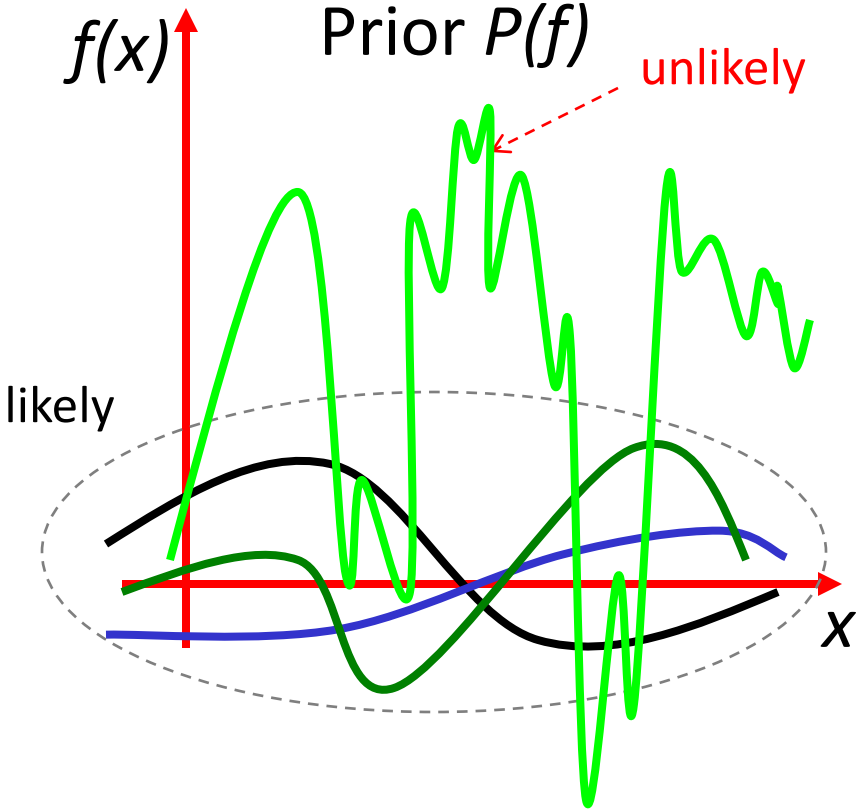


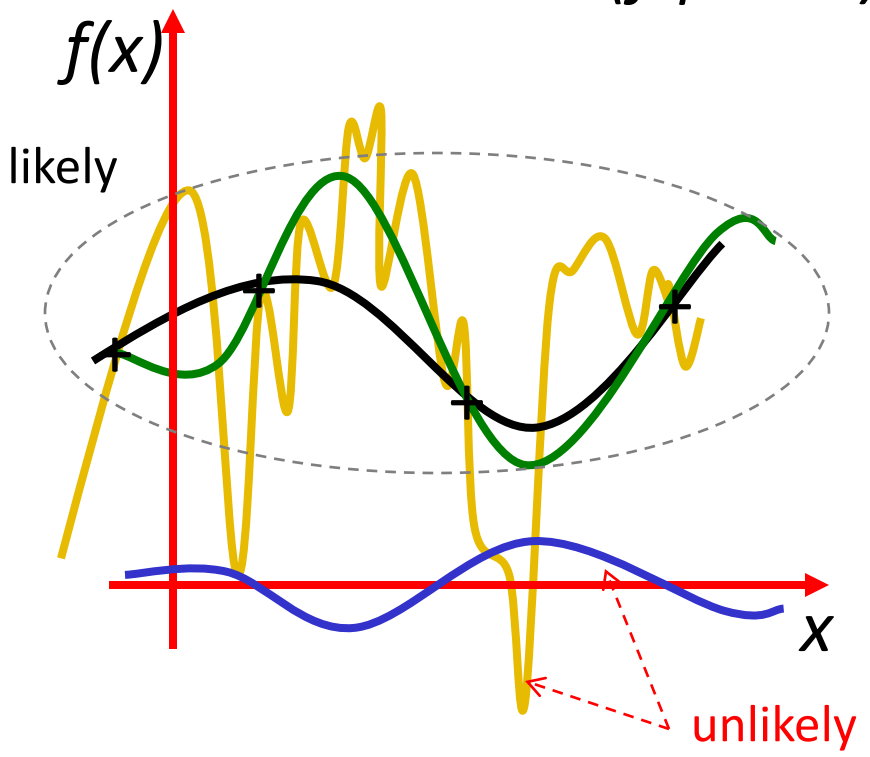
Bayesian learning with Gaussian processes

[c.f. Rasmussen & Williams 2006]

Likelihood: *P(data | f)*

➔ Posterior: *P(f | data)*





Predictive uncertainty + tractable inference

Gaussian process

A Gaussian Process (GP) is an

(infinite) set of random variables, indexed by some set X

i.e., there exists functions $\mu : X \to \mathbb{R} \quad k : X \times X \to \mathbb{R}$

and for each $x \in X$ there's a random variable f_x

such that for all $A \subseteq X$, $A = \{x_1, \ldots, x_m\}$

it holds that $f_A = [f_{x_1}, \ldots, f_{x_m}] \sim N(\mu_A, K_{AA})$

for $\mathbf{K}_{AA} = \begin{pmatrix} k(x_1, x_1) & k(x_1, x_2) & \ldots & k(x_1, x_m) \\ \vdots & & \vdots & \\ k(x_m, x_1) & k(x_m, x_2) & \ldots & k(x_m, x_m) \end{pmatrix}$ and $\mu_A = \begin{pmatrix} \mu(x_1) \\ \vdots \\ \mu(x_m) \end{pmatrix}$

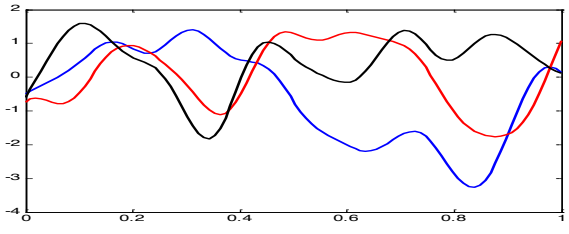
k is called covariance (kernel) function

μ is called mean function

We write: $f \sim GP(\mu, k)$ and define $f(x) := f_x$

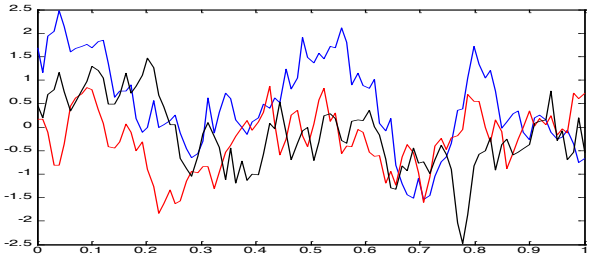
Covariance functions examples:

- Squared exponential (Gaussian / RBF):



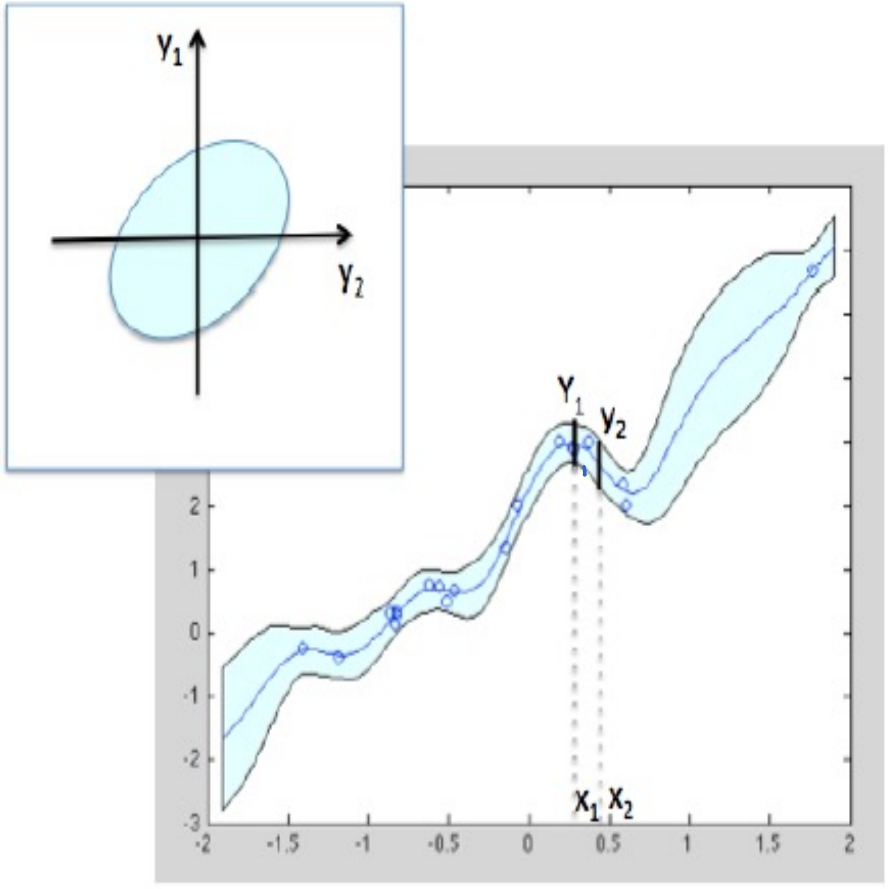
$$k(x, x') = \sigma_p^2 \exp\left(-\frac{\|x - x'\|_2^2}{2h^2}\right)$$

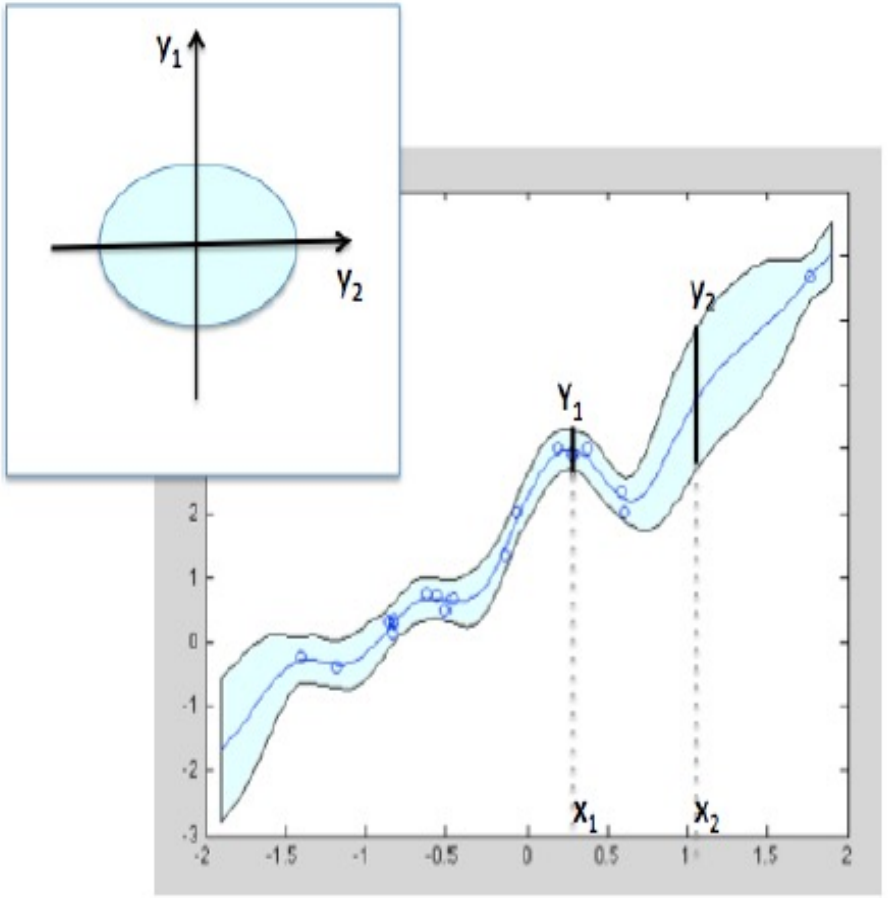
- Exponential kernel:



$$k(x, x') = \sigma_p^2 \exp\left(-\frac{\|x - x'\|_2}{h}\right)$$

Illustration of covariance





$$b(x_1, x_2) = \mathrm{Cov}(f(x_1), f(x_2)) = \mathbb{E}[(f(x_1) - \mu(x_1)) \cdot (f(x_2) - \mu(x_2))]$$

Making predictions with GPs

- Suppose $f \sim GP(\mu, k)$ and for set $A = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$

  we observe $y_i = f(\boldsymbol{x}_i) + \epsilon_i$ where $\epsilon_i \sim N(0, \sigma_n^2)$
- Then $p(f \mid \mathbf{x}_1, \ldots, \mathbf{x}_m, y_1, \ldots, y_m) = GP(f; \mu', k')$

  where

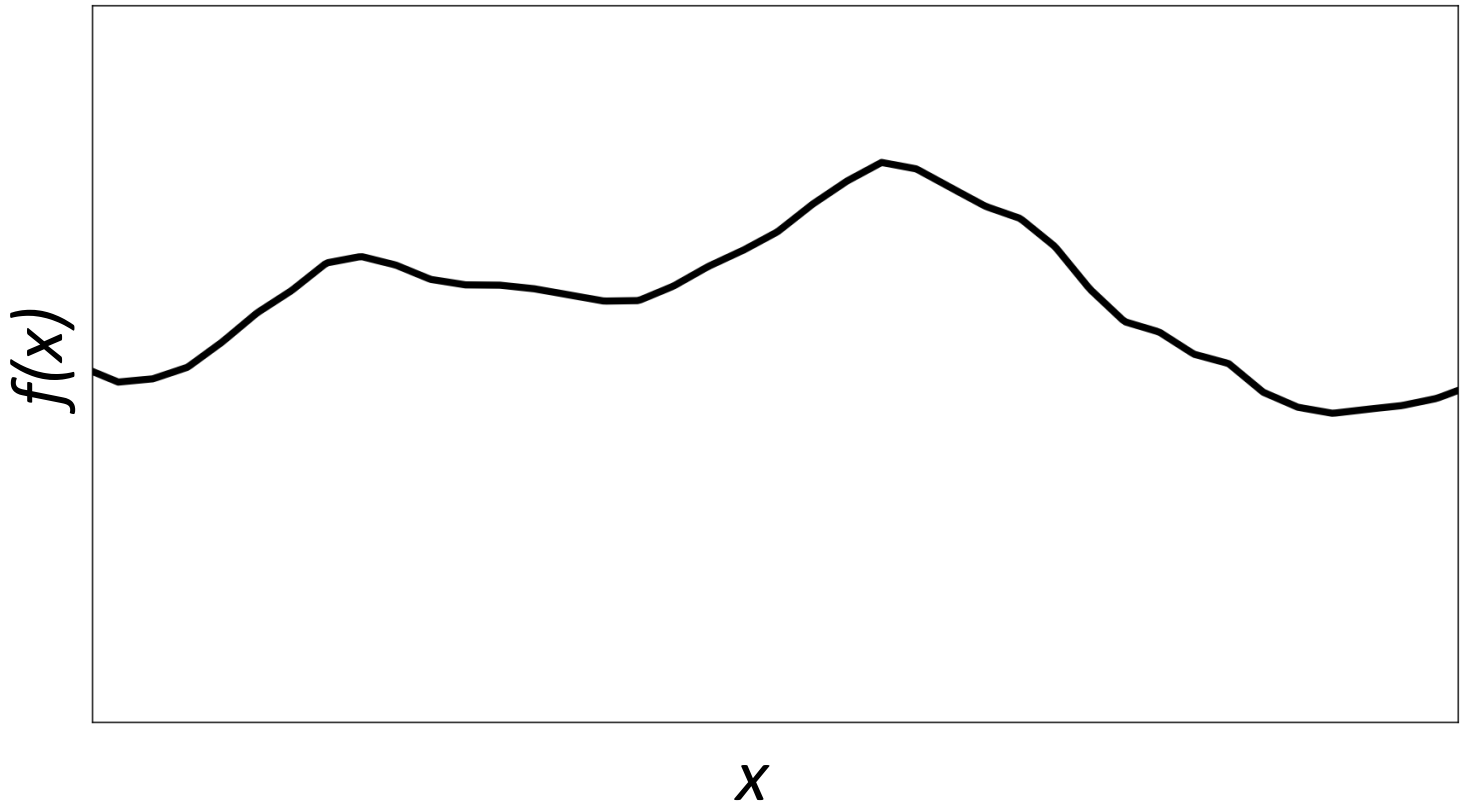
$$\mu'(x) = \mu(x) + \boldsymbol{k}_{x,A}(\boldsymbol{K}_{AA} + \sigma^{-2}\boldsymbol{I})^{-1}(\boldsymbol{y}_A - \mu_A)$$

$$k'(x, x') = k(x, x') - k_{x,A}(K_{AA} + \sigma^2 I)^{-1}k^T_{x',A}$$

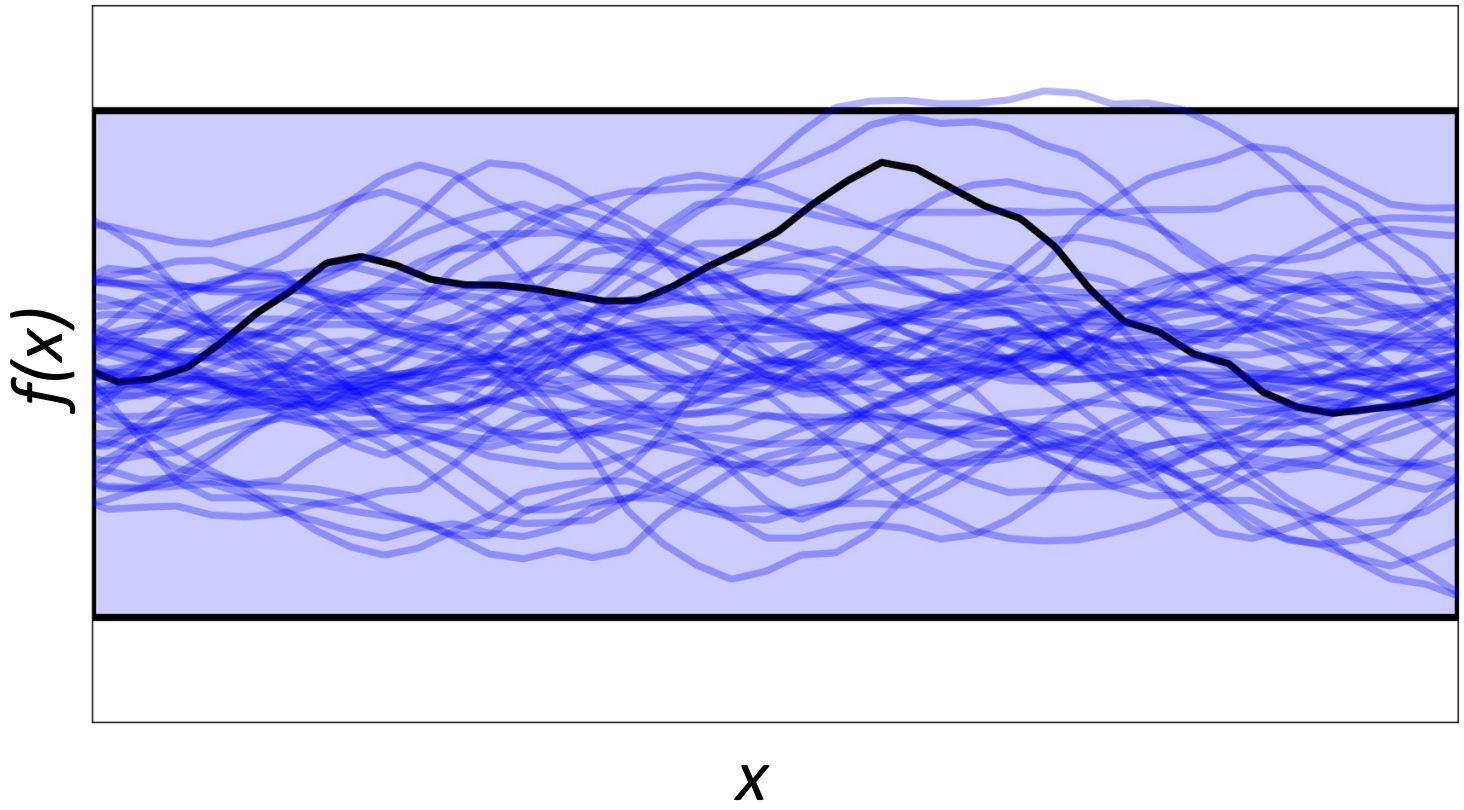
➔ Closed form formulas for prediction!

➔ Posterior covariance k' does not depend on $\boldsymbol{y}_A$

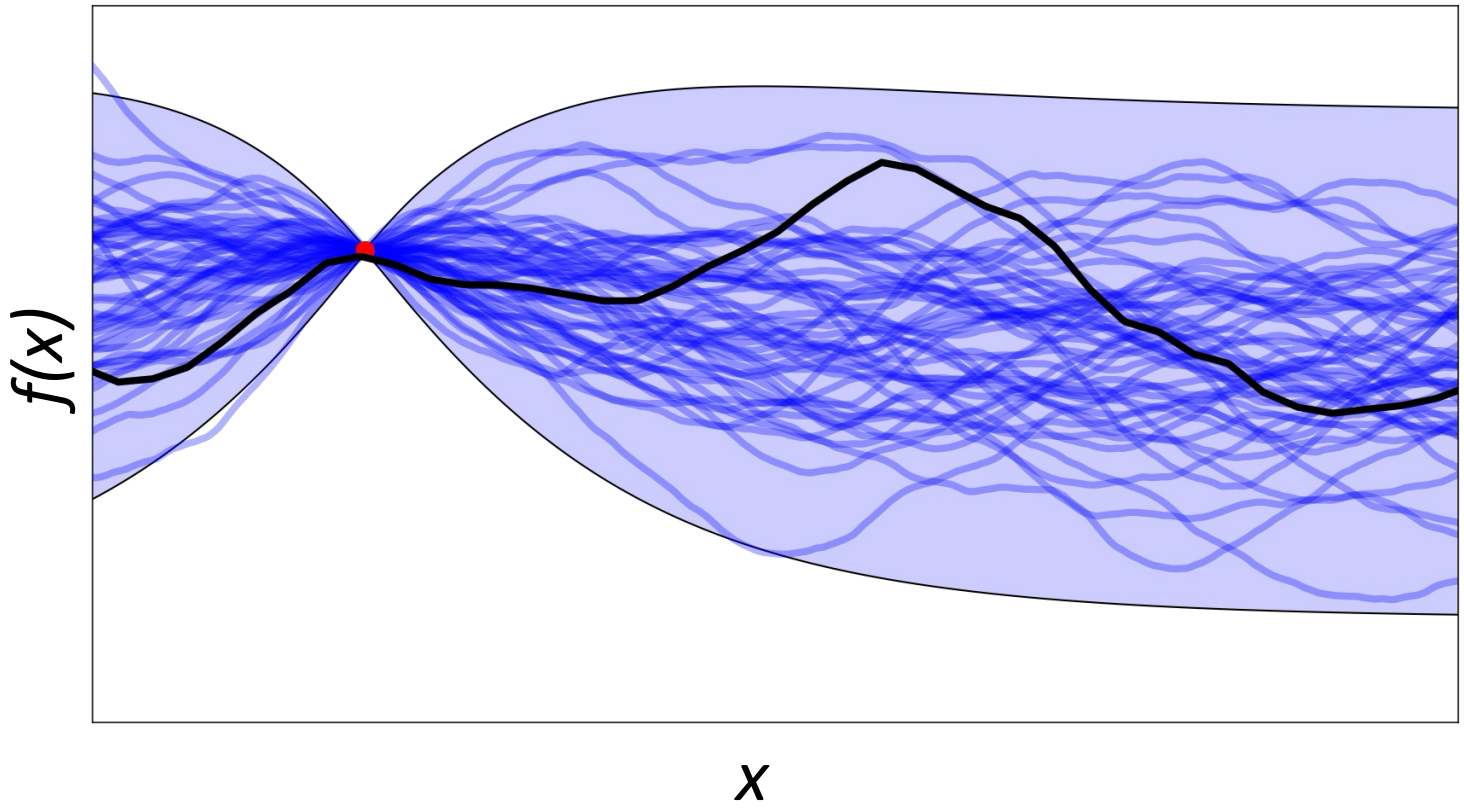
GP Inference Illustration



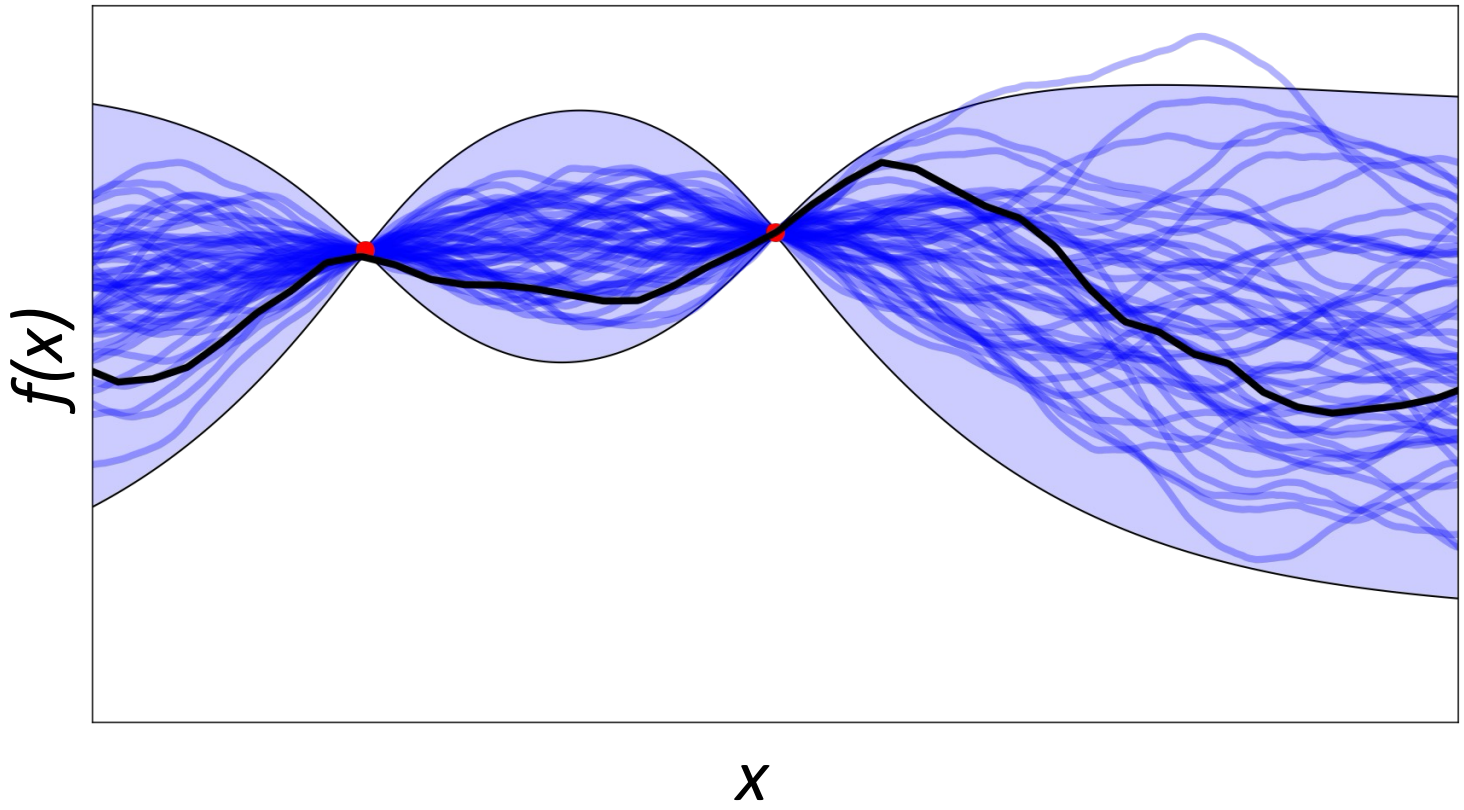
GP Inference Illustration



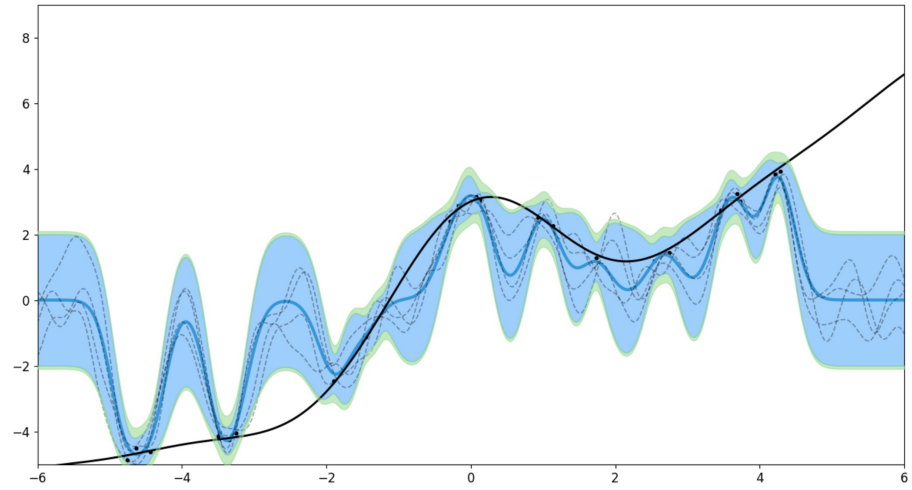
GP Inference Illustration

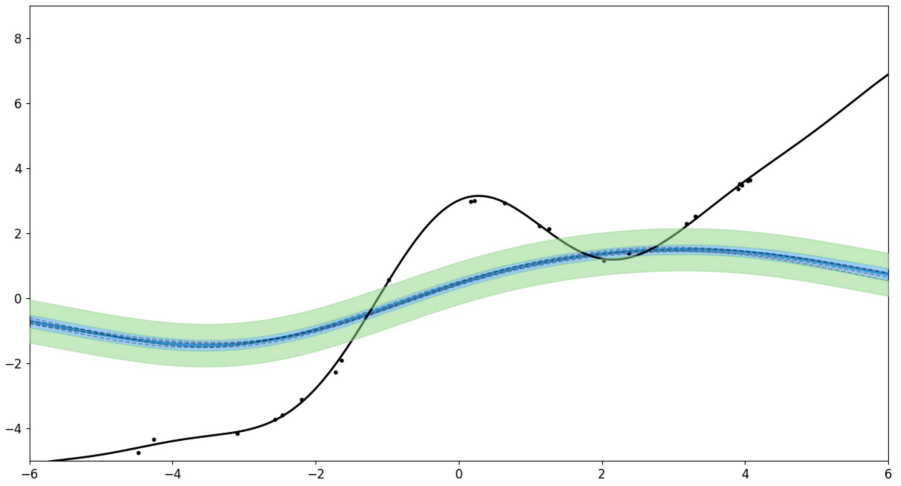


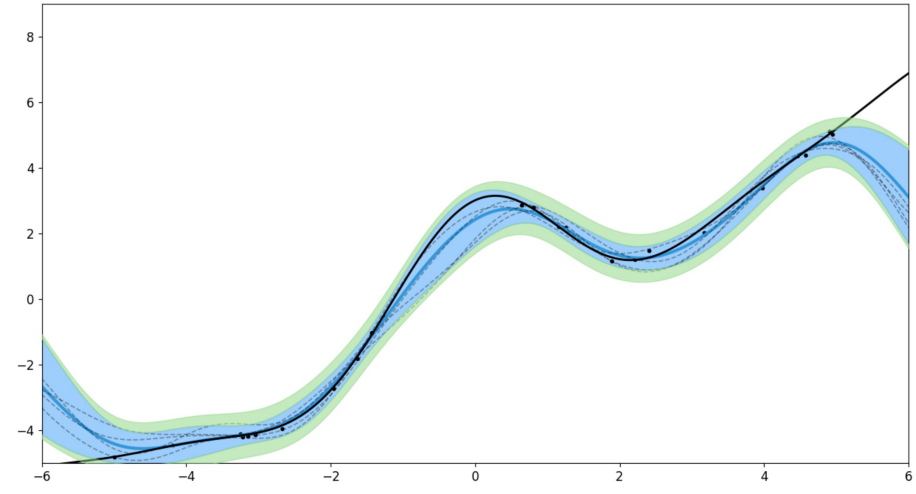
GP Inference Illustration

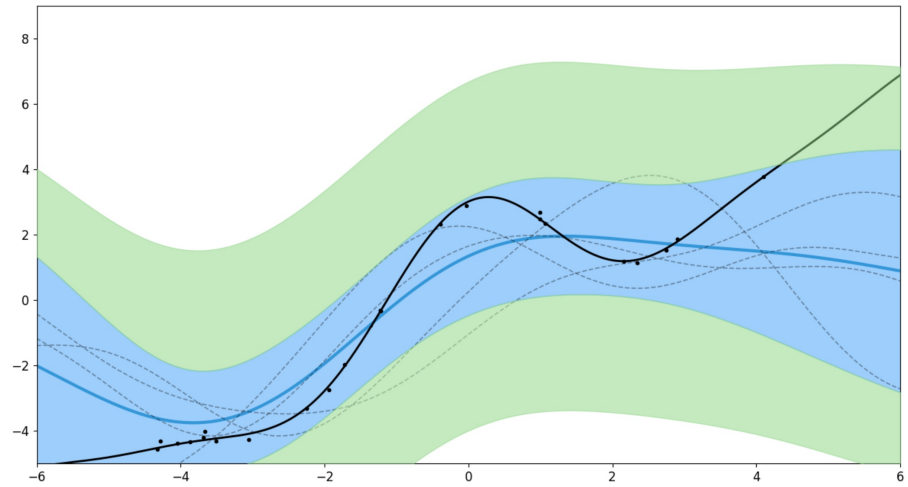


Effect of kernel parameters







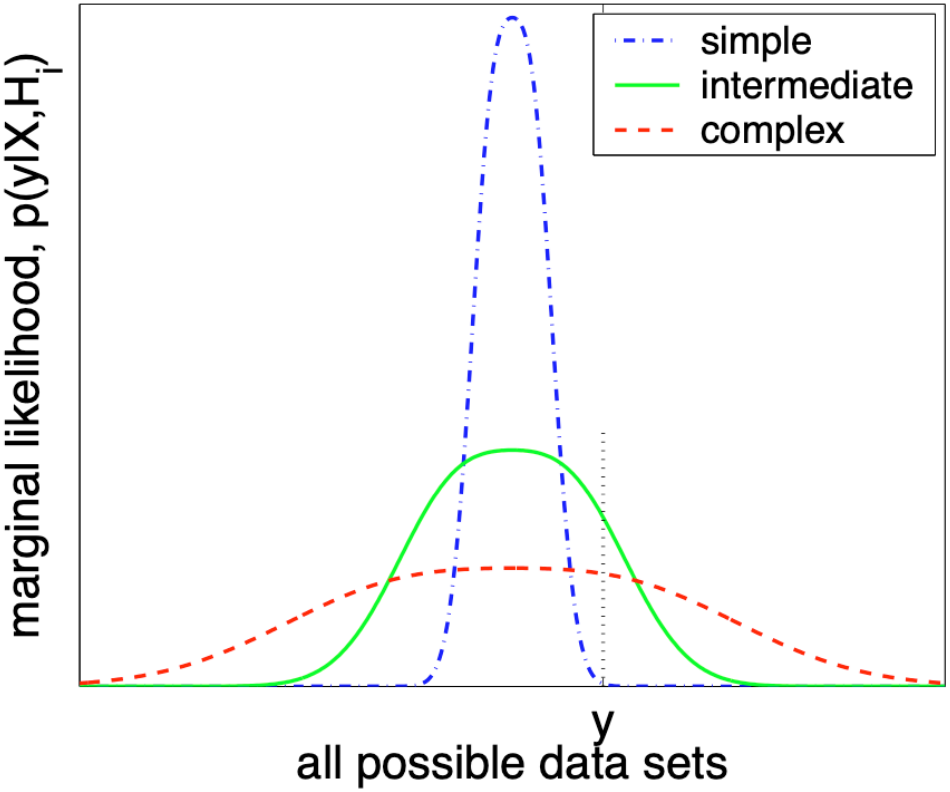


Optimizing kernel parameters

- How should we pick the hyperparameters?
- One answer: crossvalidation on predictive performance.

- The Bayesian perspective provides an alternative approach: Maximize the marginal likelihood of the data

Bayesian model selection



[Rasmussen & Williams, Gaussian Processes for ML]

Optimizing marginal likelihood for GPs

Model selection for GPs

- Marginal likelihood of the data

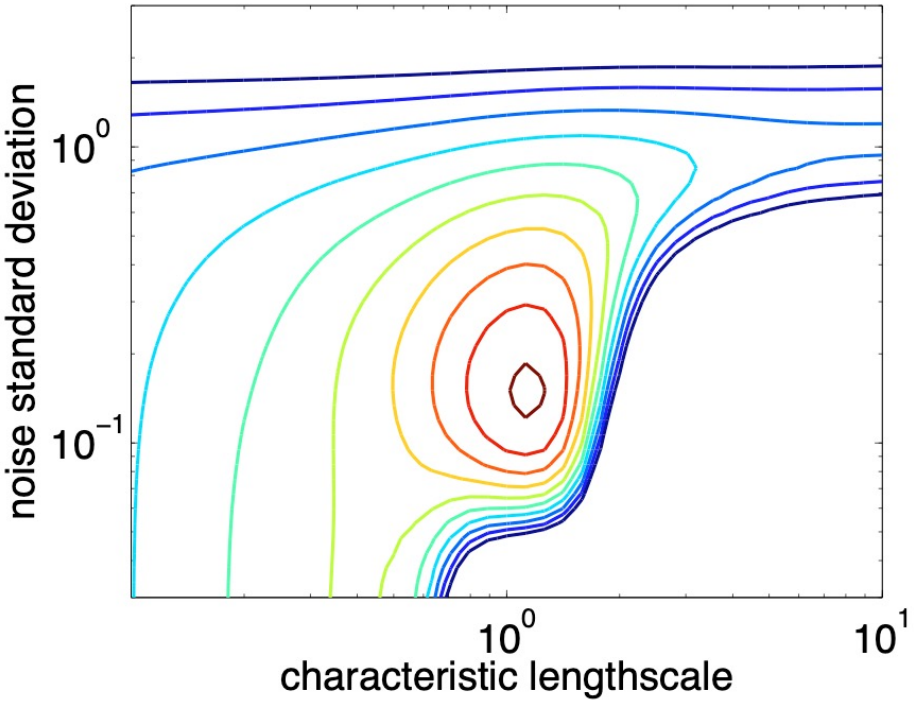
$$\log p(\, y \mid X, \theta \,) = -\frac{1}{2}\boldsymbol{y}^{\boldsymbol{T}}\left(\boldsymbol{K}_{\boldsymbol{y}}^{(\boldsymbol{\theta})}\right)^{-1}\boldsymbol{y} - \frac{1}{2}\log\left|\boldsymbol{K}_{\boldsymbol{y}}^{(\boldsymbol{\theta})}\right| - \frac{n}{2}\log 2\pi$$

Optimizing the likelihood

Gradient of the likelihood

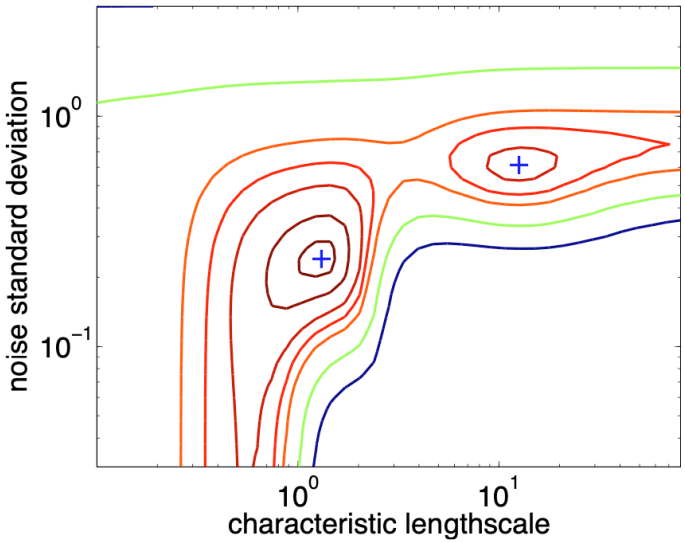
$$\frac{\partial}{\partial_{\theta_j}} \log p(\boldsymbol{y} \mid \boldsymbol{X}, \theta) = \frac{1}{2} \mathrm{tr}\left(\left(\boldsymbol{\alpha}\boldsymbol{\alpha}^{\boldsymbol{T}} - \left(\boldsymbol{K}^{(\boldsymbol{\theta})} \right)^{-1} \right) \frac{\partial \boldsymbol{K}^{(\boldsymbol{\theta})}}{\partial \theta_j} \right)$$

$$\boldsymbol{\alpha} = \left(\boldsymbol{K}^{(\boldsymbol{\theta})} \right)^{-1} \boldsymbol{y}$$

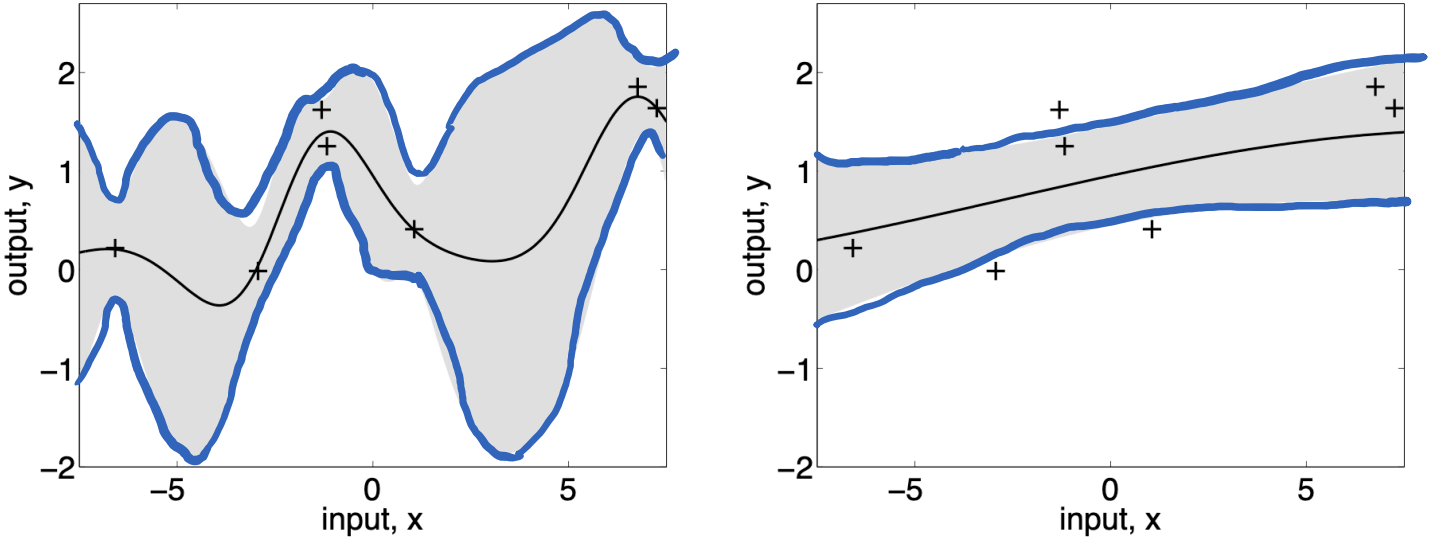


[Rasmussen & Williams GPML '06, pp 113-114]

Local optima



(a)



[Rasmussen & Williams GPML '06, pp 116]

Demo

Bayesian Model Selection

- Maximizing marginal likelihood is an example of an Empirical Bayes method – estimating a prior distribution from data
- Integrating (rather than optimizing) over the unknown function helps guarding against overfitting
- Other possibilities exist:
  - Can place hyperprior on parameters of the prior and obtain MAP estimate (corresponds to a regularization term)
  - Can integrate out the hyperprior (but also has params …)

Computational issues

Computational cost of prediction with a GP?

$$\mu'(x) = \mu(x) + \boldsymbol{k}_{x,A}(\boldsymbol{K}_{AA} + \sigma^{-2}\boldsymbol{I})^{-1}(\boldsymbol{y}_A - \mu_A)$$

$$k'(x, x') = k(x, x') - \boldsymbol{k}_{x,A}(\boldsymbol{K}_{AA} + \sigma^2 I)^{-1}\boldsymbol{k}_{x',A}^T$$

- Exact computation requires solving linear system in $|A| = n$ variables ➔ $\Theta(n^3)$
- This is in contrast to Bayesian linear regression: $\Theta(nd^2)$ (can even be maintained recursively at same cost)

Fast GP methods

- Basic approaches for acceleration:
  - Exploiting parallelism (GPU computations)
  - Kernel function approximations (RFFs, QFFs, …)
  - Local GP methods
  - Inducing point methods (SoR, FITC, VFE etc.)

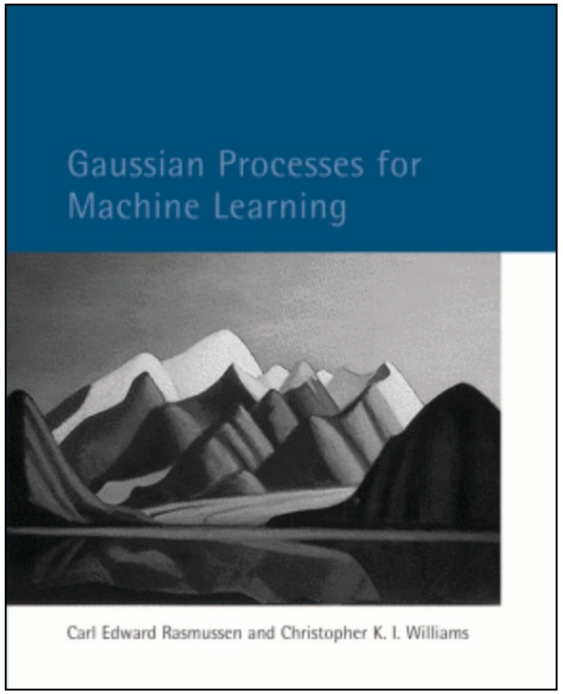
- We will revisit this challenge in context of approximate inference

Summary

- Gaussian processes = kernelized Bayesian Linear Regression
- Can compute marginals / conditionals in closed form
- Optimize hyperparameters via maximizing the marginal likelihood
- Various fast approximations to exact Gaussian process inference exist

Reading

- Rasmussen & Williams: Gaussian Processes for ML
- Chapter 2: 2.1.1-2.3
- Chapter 4: up to 4.2



- Quiñonero-Candela & Rasmussen: “A Unifying View of Sparse Approximate Gaussian Process Regression”, JMLR 2005
- Rahimi & Recht: “Random Features for Large-Scale Kernel Machines”, NeurIPS 2007